Battery State of Charge (SOC) Estimation Using Deep Learning Across Multiple Battery Chemistries

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Objective

The objective of this work was to investigate the performance of deep learning models for Battery State of Charge (SOC) estimation across different lithium-ion battery chemistries. Two recurrent neural network architectures, Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU), were implemented and evaluated on multiple benchmark battery datasets.

Datasets Used

Four publicly available battery datasets representing different battery chemistries were selected.

Dataset	Chemistry	Input Features	Target
LG HG2	NCA (Nickel Cobalt Aluminum Oxide)	Voltage, Current, Temperature	SOC
HUST	LFP (Lithium Iron Phosphate)	Voltage, Current, Temperature	SOC
NASA PCoE	Lithium-Ion	Voltage, Current, Temperature	SOC
Tongji EV Battery Dataset	NMC (Nickel Manganese Cobalt)	Voltage, Current	SOC

Methodology

The following workflow was applied to all datasets:

1. Dataset acquisition and preprocessing
2. Feature selection and SOC target preparation
3. Data normalization using Min-Max Scaling
4. Sequence generation with a timestep length of 20
5. Train-test split
6. Model training using:
    - Long Short-Term Memory (LSTM)
    - Gated Recurrent Unit (GRU)
7. Performance evaluation using:
    - Loss (MSE)
    - Mean Absolute Error (MAE)
    - Root Mean Square Error (RMSE)

Experimental Results

Dataset	Chemistry	LSTM MAE	GRU MAE	Best Model
LG HG2	NCA	1.47	1.38	GRU
HUST	LFP	0.0154	0.0207	LSTM
NASA PCoE	Li-ion	0.1926	0.1829	GRU
Tongji EV	NMC	0.2292	0.2236	GRU

Cross-Dataset Analysis

The experimental results indicate that model performance varies across battery chemistries and dataset characteristics.

- GRU achieved superior performance on three datasets (NCA, NASA Li-ion, and NMC).
- LSTM achieved the best performance on the LFP-based HUST dataset.
- The NASA and Tongji datasets exhibited higher prediction errors due to larger variability, multiple operating conditions, and heterogeneous battery behavior.
- HUST produced the lowest error values, indicating a comparatively cleaner and more consistent battery dataset.

- GRU demonstrated stronger overall generalization capability across multiple battery chemistries while maintaining a simpler architecture.

Conclusion

A comparative SOC estimation study was successfully conducted on four battery chemistries using LSTM and GRU neural networks. The results demonstrate that recurrent neural networks can effectively estimate SOC from battery operational measurements. While both architectures performed well, GRU achieved the highest number of dataset-specific wins and exhibited better overall generalization across different battery chemistries.

This study highlights the importance of evaluating SOC estimation models across diverse battery datasets, as model effectiveness depends not only on network architecture but also on battery chemistry, operating conditions, and dataset complexity.